HW4 - Network Visualizations

Name:_____

Deadline 4 December 2022

Network Visualizations

Problem 1 (70 points)

1) Load the file "transfers.csv" into a dataset called **transfers**.
   Group by the variables **League_from** and **League_to**, then create new variables **Mean_Age**(Mean of the variable **Age**), **Mean_Fee**(Mean of the variable **Transfer_fee** divided by 1000000) and **Count**(number of transfers in that group) by using the summarize function and filter out the observations with less than 10 transfers in between (**Count** >= 10) (5 points)

2) Create an igraph using `graph_from_edgelist` for the dataset you just got. (5 points)

3) Set edge attributes for the summarized three variables and use their values. (5 points)

4) Draw the network in a circle layout, pick one of the attributes as the width of the edge.
   Were you able to get an interpretable graph ? If no what's the reason ? If yes intepret it :) (10 points)

5) Apply a min-max or any other reasonable transformation to the variable chosen by you and visualize again using the scaled variable for now. !Hint, if you are using min/max replace 0 with a small number. (10 points)

6) Interpret the graph (10 points)

You can write your interpretations directly here:_____

7) Using the dataset *transfers* create another subset of data. For now, pick any of the leageus that you want from the available ones and filter out the internal transfers in that leageue (meaning that the variables *League_from* and *League_to* are equal to each other). Now group by the variables *Team_from* and *Team_to* and summarize the same values as in Problem 1. (5 points)

8) Create an igraph network and set the edge attributes for the graph, the same way as in problem 3. (5 points)

9) Now use that graph and draw a chordDiagram and use any of the three atributes you desire, apply the transformations to the variables if needed. (10 points)

10) Interpret the visualization. (5 points)

Network Statistics (30 points)

Problem 2

1) Using the network created in **Problem 1.2** calculate the edge density of the network. Interpret the result. (5 points)

2) Now calculate and interpret the reciprocity of the network. (5 points)

3) Find the most active league of the market. by calculating the degree centrality measure (5 points)

4) Calculate the closeness statistics of the network, pick any of the edge attributes as a weight and interpret the results. (5 points)

5) Now calculate the page rank of the network, which league is the most popular ?. (5 points)

6) Detect communities in the network. (5 points)